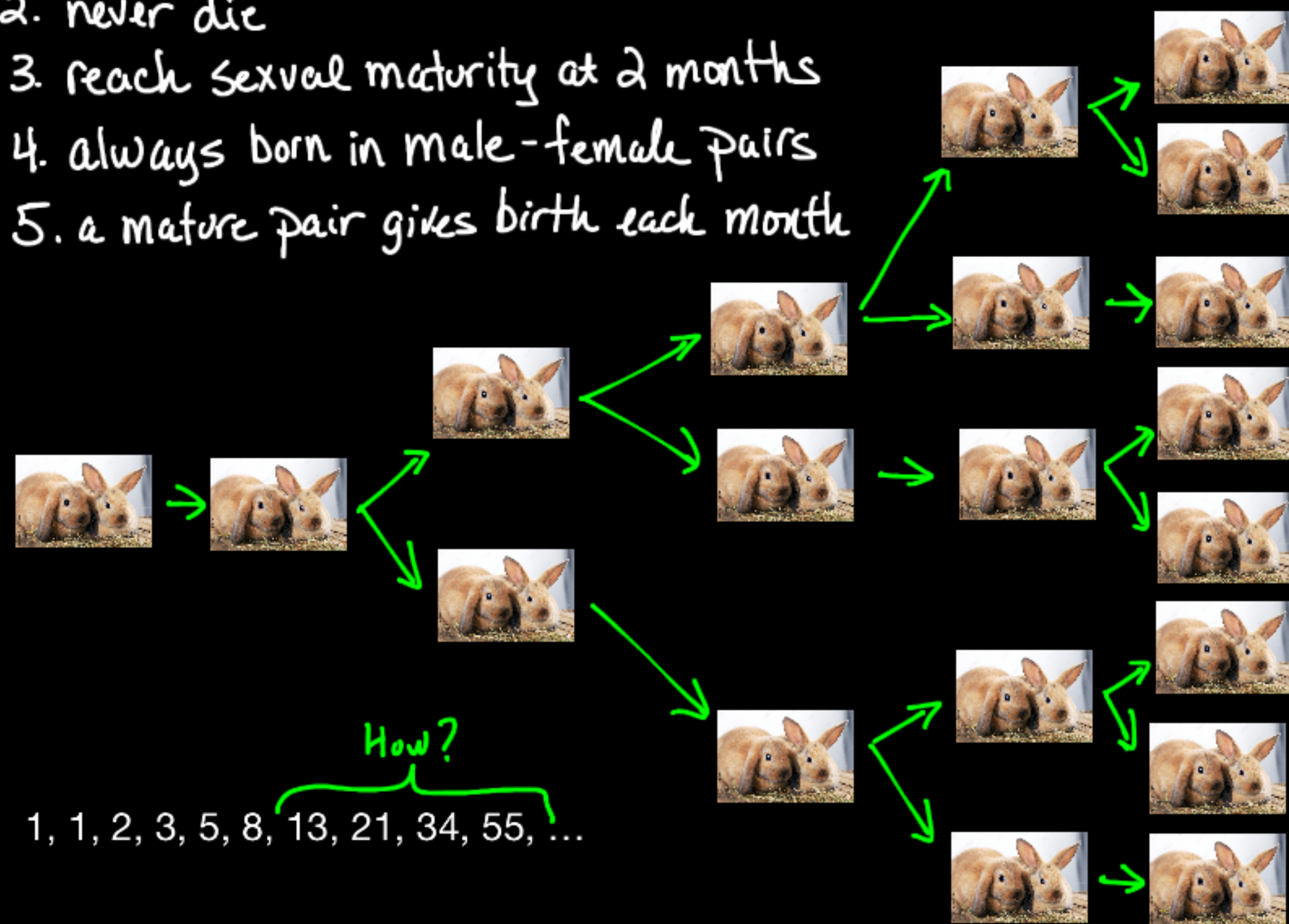
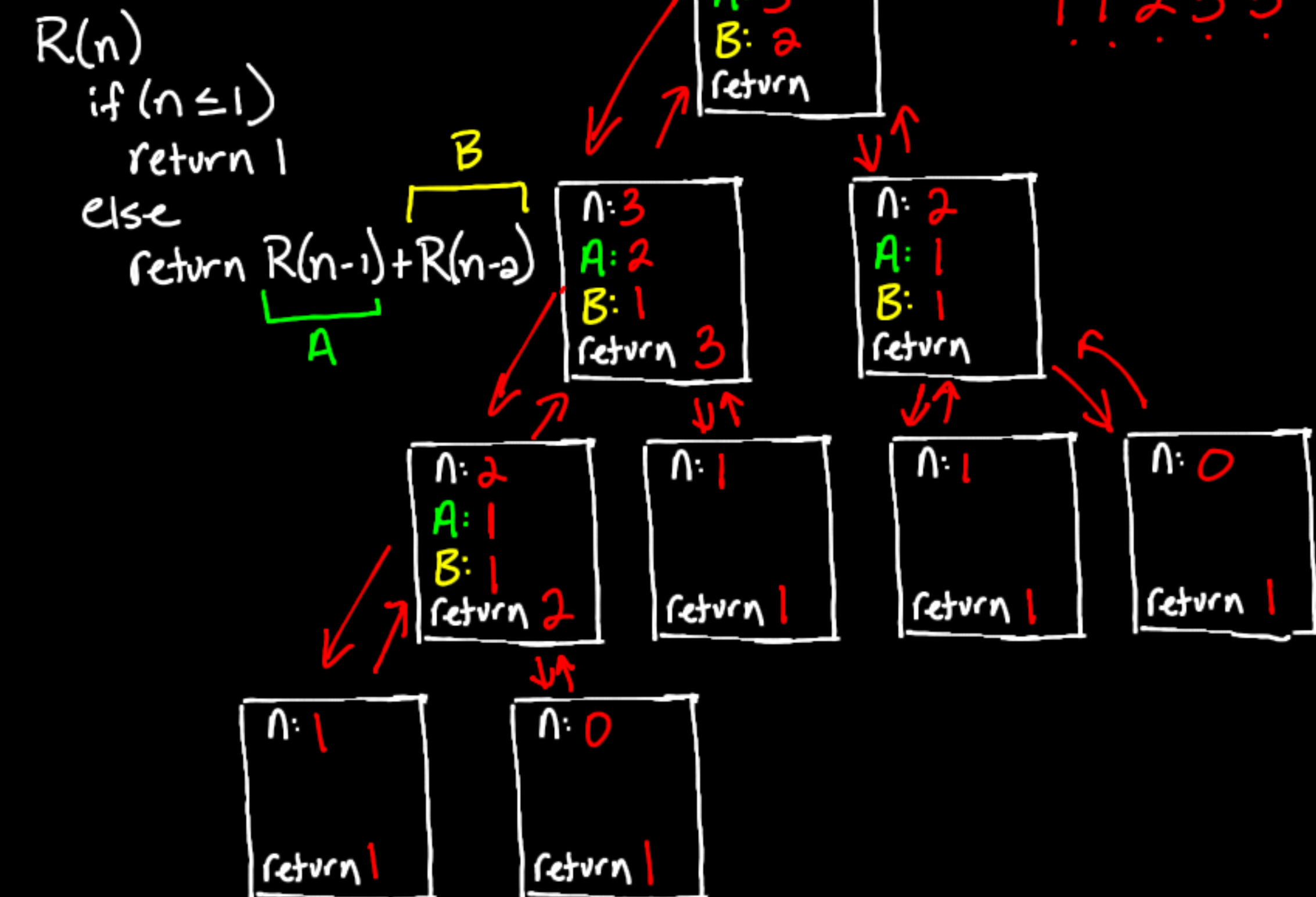


① "Alternative facts" about rabbits

1. prolific breeders
2. never die
3. reach sexual maturity at 2 months
4. always born in male-female pairs
5. a mature pair gives birth each month



③ Box Trace



② Array w/ arithmetic

	0	1	2	3	4	5	6	7	8	9	
R	1	1	2	3	5	8	13	21	34	55	...

$R[0] = 1$
 $R[1] = 1$
 $R[2] = R[0] + R[1]$
 $R[3] = R[1] + R[2]$
 $\vdots$

Iteration

pprev = prev = 1
for ($i=2, i \leq n; i++$)
 $R = \text{pprev} + \text{prev}$
 $\text{pprev} = \text{prev}$
 $\text{prev} = R$

Iteration values:
 $i=2$: $R=2$
 $i=3$: $R=3$
 $i=4$: $R=5$
 $i=5$: $R=8$
 $\vdots$

Recursive

$R(5) = R(3) + R(4)$
 $R(4) = R(2) + R(3)$
 $R(3) = R(1) + R(2)$
 $R(2) = R(0) + R(1)$
 $R(1) = 1$
 $R(0) = 1$

$R(0) = 1$

$R(1) = 1$

$R(n) = R(n-2) + R(n-1)$

Recurrence Relation

$R(n) = \begin{cases} 1 & n \leq 1 \\ R(n-2) + R(n-1) & n > 1 \end{cases}$

④ Recursion Tree

